

Engineering Innovation Challenge 2023

Flood-Proof Roads

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Abstract

Floods have become a challenge in many built-up areas due to increased urban development and uncertain climate conditions. Singapore, with its low-lying geography and heavy rainfall, experiences regular flash flooding. This issue correspondingly exists globally, posing significant risks in casualties and infrastructural damages. A solution of flood-proof roads was explored to replace traditional roads in flood-prone areas, particularly those in proximity to essential facilities such as hospitals.



Figure 1: An ambulance is seen stuck in flood water near Changi fire station in Singapore on June 23, 2020 [1].

The proposed system involves the utilisation of multiple interconnected floating pontoon units beneath permeable road surfaces. During floods, the road surface and the pontoon units will employ natural buoyancy forces, supplemented by a motorised deployment system, to create a functional floating commuting surface above the flood waters. To ensure reliability, several key aspects must be addressed, including compliance with national standards for structural integrity, buoyancy and stability, material selection, adaptability to flood conditions, and overall safety and user experience.

A prototype will be designed that utilises a variety of materials and incorporates a capacitive proximity sensor for water level detection. Comprehensive testing will be conducted across

various scenarios to validate its effectiveness. These will include commutability during normal dry conditions; commutability during flood conditions from low, medium and high rainfall events; and the compatibility of the system with different types of vehicles varying in size and weight.

The flood-proof road system aims to mitigate the impact of floods on transportation infrastructure in flood-prone areas. This pitch contributes to the development of resilient urban infrastructure that is applicable in various urban settings.

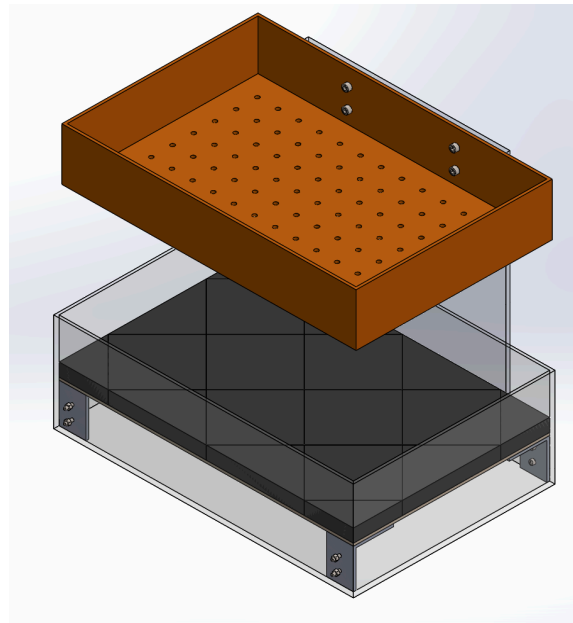


Figure 2: Small scale prototype design.

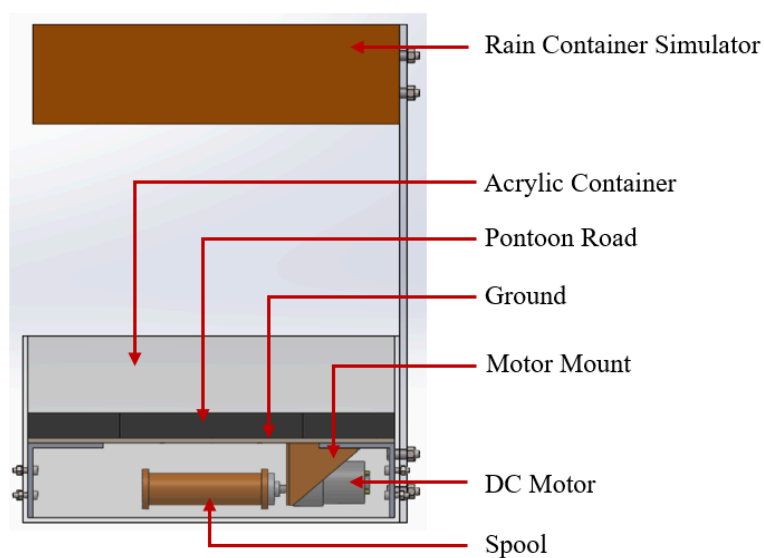


Figure 3: Major components of the prototype.

Project Proposal

Research Problem

Climate change is impacting all parts of the world in many ways. In Singapore, the observed effects include the annual rainfall rising at a rate of 101 mm per decade from 1980 to 2016, and the sea level rising at the rate of 1.2 mm to 1.7 mm per year between 1975 to 2009 [2]. It is projected that the intensity and frequency of heavy rainfall will increase, and sea levels in Singapore will rise by up to 1 m by the end of the century [2].

The rise in rainfall and sea levels mean greater risks of flooding, including river floods, flash floods and coastal floods. At its current state, Singapore receives 100 to 300 mm of rain each month, with the average annual rainfall being 2,340 mm [3], and one-third of the country is less than 5 m above sea level [4]. Hence, to build resilience and mitigate the anticipated worsening of floods in the coming years, it is essential for Singapore to proactively future-proof its city from flooding.

Prototype Proposal

To help address the flooding concerns, especially to flash flooding which is expected to worsen with rising rainfall rates, an innovative alternative solution to traditional roads is being proposed. The design concept involves having pontoons installed under porous roads that can automatically detect the rise in water levels and raise the level of roads so that they are above the water when flash flooding occurs.

The prototype design aims to model a pontoon road system that can deploy quickly in time of flash flooding, with sufficient buoyancy for an increasing demand of traffic. The structure consists of interconnected pontoon pads made primarily of wood, the deployment system controlled by string-spools with attached DC motors and capacitive proximity sensors for water level detecting. 3D printing is used for rapid prototyping the rain container and the system mount, in which the integrated system is validated under three simulated scenarios to assess (1) structural integrity and strength under vehicle loads (2) water flow and system response, and (3) performance with varying loads.

For a more detailed analysis of the prototype proposal, please refer to the Appendix.

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Appendix

Current Flooding Solutions Available

Current solutions that Singapore has implemented include green roofs, retention ponds, detention tanks, porous pavements, rain gardens, diversion canals, canal improvements, minimum platform and crest levels, reservoirs, and flood barriers [5]. Although these solutions have helped, they have not effectively eliminated flooding from Singapore, with floods still occurring in Singapore. Furthermore, these floods are dangerous and can significantly impact the operations of a city and the lives of its people. For example, road closures resulting from flooding can prevent ambulances from delivering patients to hospitals in a timely manner, as well as individuals may take the risk of driving through flood waters, even though it is unsafe to do so. Moreover, flooding is a major concern not only in Singapore but also in many parts of the world and as a result, a solution would not only benefit Singapore but also numerous cities worldwide.

Project Objectives

Research Question

What innovative approach can be utilised to develop a resilient solution that enables safe commuting for drivers on flood-prone roads?

Aim

The overall aim of the project is to create a solution that will allow drivers to safely commute over roads that would usually be closed due to flooding. It is hoped that the solution could be implemented instead of traditional asphalt roads, especially on roads that are usually prone to flooding as well as critical roads such as roads near hospitals.

Hypothesis

It is hypothesised that incorporating pontoons underneath porous roads would allow for water to seep through while enabling the roads to float above, which ultimately facilitates a safe commuting of drivers on roads that are commonly closed during flooding incidents.

Engineering Goals

The engineering goals that need to be achieved in order to produce a viable solution include:

1. *Structural Integrity:* Ensure that the pontoon road design can withstand the anticipated loads and stresses, including the weight of vehicles, water pressure, and dynamic forces during floods.
2. *Buoyancy and Stability:* Design the pontoons to provide sufficient buoyancy to keep the road afloat during flood events, maintaining stability and preventing any tilting or instability.
3. *Material Selection:* Identify appropriate materials for the pontoon structure and porous road surface that can withstand water exposure, provide sufficient strength, and ensure durability over the expected lifespan of the road.
4. *Adaptability to Flood Conditions:* Design the pontoon road system to be responsive and adaptable to varying flood conditions, including water levels, flow rates, and duration, to maintain functionality and safety.
5. *Safety and User Experience:* Prioritise the safety of drivers, pedestrians, and other road users by incorporating features that enhance visibility, provide adequate road grip, and mitigate potential hazards associated with the pontoon road design.

Expected Outcomes

On completion of the project, expected outcomes include:

1. *Scalable and Adaptable:* The goal is to build an accurate small-scale working prototype of a section of a road pontoon. It is expected that this pontoon road prototype is adaptable and scalable to fit various flood-prone areas and road networks. The expected outcome is a flexible solution that can be customised based on the specific requirements and conditions of different locations.
2. *Normal Commutability during No-Flood Periods:* As this solution is intended to be in place at all times, even when flooding does not occur, the prototype is expected to demonstrate that it is functional under normal, non-flood conditions.
3. *Enhanced Commutability during Floods:* The prototype is expected to demonstrate its ability to prevent flooding on roads by allowing water to seep below the road surface and safely raising the roads when flooding occurs. This is expected to allow

small-scaled cars to traverse the roads safely. Furthermore, it is anticipated that as water levels lower, the roads will gradually lower as well.

Prototype Design

The main goal for this system is to deploy the pontoon structures in time according to the flash flood conditions, while exhibiting enough buoyancy forces to allow increasing traffic demands.

Pontoon road

This structure consists of multiple pontoon pads, interconnected with each other and all attached by motorised deployment system as shown in Figure 4, and can be overlaid with additional layers (asphalt for e.g.). The appropriate materials and pattern designs are essentials for each pad to create a permeable yet long-lasting pontoon.

In today's world, floating bridges using pontoons are usually made out of a fusion of concrete, steel and wood with long-term corrosion and damping effects protection due to traffic, wind and waves [6]. To incorporate these, wood is chosen for the main materials of the proposed prototype as it is cost-effective and easy to fabricate.

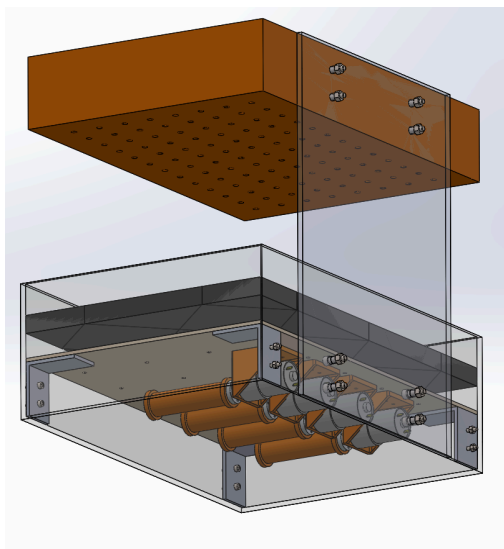


Figure 4: Prototype design made in SolidWorks.

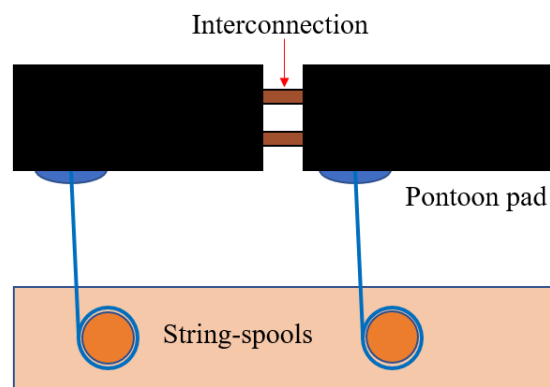


Figure 5: Prototype design illustrated side view.

Pontoon deployment system

The deployment is controlled via a system of string-spools with attached DC motors as shown in Figure 5, which allows the free floating and self-deployment of the pontoons under buoyancy forces. However, under prototyping scale, a capacitive proximity sensor can be used to detect the rising water level, thus, send signals to the motor for deploying the pontoon road accordingly. This type of sensor is used for sensing/measuring target's position in a wide variety of manufacturing and automation applications, which relies on the charging property of the capacitors [7].

From here, the spool and the mounting base are 3D printed to allow quick prototyping, using well-known PLA material, a user-friendly thermoplastic with high strength but low melting temperature. The design needs to be optimised so that in the implementation phase, there will be minimal modification/reconstruction on the pre-existing road structures, with ease of instalment and maintenance afterwards.

Rain simulating system

To validate the design accurately, a 3D printed container with the same base area as the pontoon road is mounted on top to simulate the rain. The container is designed with multiple holes with variable sizes to adjust the rainfall rate, thus, different raining scenarios can be tested and evaluated.

Lastly, the whole system is contained in a clear acrylic box to replicate the flooding condition, providing a clear view of the water level and the deployment system.

Prototype Cost Analysis

A high level breakdown of the expected cost for creating the prototype can be found in Table 1. This is only an initial estimate and costs may vary depending on the successfulness of the prototype. Note that miscellaneous electrical components as well as the general tools that will be used are fortunately provided by Monash University, Australia free of charge.

Table 1: Cost Breakdown.

Product	Quantity	Total Cost (AUD)	Source
DC motor (6 VDC rated)	4	\$8	[8]
PLA filament roll	1	\$25	[9]
7mm Plywood	0.06 m ²	\$3	[10]
5mm Transparent Acrylic Board	0.2 m ²	\$33	[11]
Adhesive acrylic glue	1	\$12	[12]
Capacitive proximity sensor	1	\$4	[13]
Arduino Nano	1	\$20	[14]
High strength braided fishing line	50m	\$25	[15]
Miscellaneous electrical components	~	~	Monash Makerspace
Estimated Total Prototype Cost		\$130	

Testing Methodology

To validate the proposed design, three main scenarios are to be conducted on the small-scale prototype in order to confirm the feasibility of the solution.

Scenario 1: Commutability during Dry Conditions

It is necessary to test the small-scale prototype under dry conditions, without any flooding or rain, as the roads are expected to be functional in all weather conditions. The structural integrity of the pontoon roads, as well as the selected materials used in their construction, need to be validated. To achieve this, the performance of the road under small-scale vehicle loads should be tested to determine its strength and ability to withstand the loads. For this testing, three vehicles need to be appropriately scaled (dimensions and weight) and tested incrementally. The three vehicles to be scaled appropriately include Light Goods Vehicles (with a maximum weight of 3,500kg), Heavy Goods Vehicles (with a maximum weight of 16,000kg), and Very Heavy Goods Vehicles (with a weight exceeding 16,000kg; for testing purposes, a weight of 30,000kg will be assumed as the maximum) [16]. If the road successfully withstands all three vehicle types, additional weights should be added until the

material deforms, in order to determine the weight at which the structure yields. Furthermore, finite element analysis (FEA) should be performed to validate the results. If the road deforms under a certain vehicle type, it is important to consider whether redesigning the pontoon will help, or alternatively, to consider implementing road restrictions such as signs indicating 'No Heavy Trucks'.

Scenario 2: Commutability during Low, Medium and High Rainfall (Flash Flooding)

After conducting testing under dry conditions, the prototype is ready to be tested under wet conditions. Starting with testing for light and medium rainfall, the objective is to determine whether the material selection and the design of the pontoons allow water to flow downward. As the amount of rainfall increases, the aim is to verify whether the system accurately detects the rising water level and raises the roads, while ensuring that the water continues to flow below the road surface. Once the maximum rainfall level is tested, the system should also be tested in reverse, meaning that the response of the system as the water level beneath the roads decreases should be evaluated. To conduct this testing, a scaled amount of water should be poured into the rain bucket positioned above the road. To increase the amount of rainfall, more water should be added. Furthermore, potentially considering different rainfall buckets with different flow rates can be beneficial. To assess the system's performance in reverse, there should be a pipe located underneath that can be unplugged to allow water to exit the system. As the water leaves at a certain low flow rate, the road levels should adjust accordingly. If time allows it, it would be beneficial to vary the flow rates to determine how effective the solution is and how its effectiveness changes accordingly.

Scenario 3: Commutability during Low, Medium and High Rainfall (Flash Flooding) with Different Vehicles

Similar testing to scenario 2 should be performed, but this time incorporating the three small-scale vehicles. The different scaled vehicles should be tested under the different rainfall conditions, starting and stopping while on the path, to identify any significant issues. As the amount of rainfall increases to high levels, the aim is to verify whether there is sufficient surface traction, whether the buoyancy forces of the pontoons can support the weight of the

different vehicles, and whether water begins to flow downward. Regarding buoyancy, Archimedes' principle states that the buoyant force can be calculated using Equation 1 [17].

$$F_b = - \rho g V \quad (1)$$

Where F_b is the buoyant force, ρ is the fluid density, g is the gravitational acceleration and V is the fluid volume. In order for the car and the pontoon road to float, the sum of weight forces should balance out with the buoyant force, or in other words the sum of the forces in the y-direction should be zero as outlined in Equation 2.

$$\sum F_y = 0 \quad (2)$$

Performing these calculations will be essential to making sure that there is at least enough buoyant force in the pontoons, otherwise, they will need to be redesigned. Furthermore, it is important to make sure that as the car drives over the road it does not cause the road to tip. Mathematically, this can be represented by stating that the moment needs to be close to zero, as outlined in Equation 3.

$$\sum M \approx 0 \quad (3)$$

Redesign of the pontoons may be necessary in order to minimise the potential moments caused by the vehicles driving over the road.

Risk Management Plan

The Risk Matrix that will be used can be seen in Table 2.

Table 2: Risk Matrix [18].

Likelihood	Consequences				
	Insignificant <i>Risk is easily mitigated by normal day to day process</i>	Minor <i>Delay up to 10% of Schedule. Additional cost to 10% of Budget</i>	Moderate <i>Delays up to 30% of Schedule Additional cost up to 30% of Budget</i>	Major <i>Delays up to 50% of Schedule Additional cost up to 50% of Budget</i>	Catastrophic <i>Project Abandoned</i>
Certain >90% chance	High	High	Extreme	Extreme	Extreme
Likely 50%-90% chance	Moderate	High	High	Extreme	Extreme
Moderate 10%-50% chance	Low	Moderate	High	Extreme	Extreme
Unlikely 3%-10% chance	Low	Low	Moderate	High	Extreme
Rare <3% chance	Low	Low	Moderate	High	High

A risk management plan that is considered when undertaking this project can be seen in Table 3.

Table 3: Risk Management Plan.

Risk	Likelihood	Consequences	Risk Level	Mitigation
Fires and Burns from 3D printing and soldering	Moderate	Minor	Moderate	Adequate ventilation, have a fire extinguisher on standby, use PPE
Electrical Shock from circuitry	Unlikely	Moderate	Moderate	Disconnect power before touching, use PPE, insulate any exposed wires
Toxic Fumes from Soldering	Moderate	Minor	Moderate	Adequate ventilation, use PPE, avoid unnecessary idle time and take breaks
Water Damage	Moderate	Moderate	High	Ensure waterproof enclosures, proper cable management, proper insulation and sealing